

EDUCATION

Velloore Institute of Technology <i>Bachelor of Technology in Computer Science Engineering</i>	Vellore, TN <i>Aug. 2021 – May 2025</i>
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EXPERIENCE

Systems Engineer - Big Data Engineer <i>Tata Consultancy Services</i> 5+ Data Pipelines Developed and Maintained- developed on Scala and Apache Spark for processing financial transaction. 2 AI Driven Data Validation Solutions- built and integrated ML libraries to pipelines for validation Config based SQL Execution- for fast execution using Spark DataFrame over existing Spark SQL reducing redundancy and improving readability, maintainability, testability, and integration capabilities of code.	Jul 2025 – Present <i>Bengaluru, KA</i>
Data Analyst Intern <i>Remote Software Solutions</i> - Learned and applied database design and management fundamentals, building relational schemas in PostgreSQL and writing SQL queries for data extraction and transformation. - Worked with the analytics team on a client project, building 3+ tiles on a dashboard in Power BI to support business insights and decision-making.	Aug. 2023 – Sep. 2023 <i>Porvorim, GA</i>

PROJECTS

LLM Fine-Tuning: Preference-Based Response Classifier - Orchestrated a pipeline in PyTorch to train a RoBERTa transformer on pairwise LLM preference data with 50K+ records - Implemented data preprocessing, prompt-response formatting, tokenization, hyperparameter tuning, training, and evaluation; experimented with multiple hyperparameter configurations to optimize classification performance. - Explored and gained foundational knowledge and depth in preference-based LLM alignment techniques including Direct Preference Optimization (DPO), RLHF, RLAIIF, LoRA, and QLoRA Tech: Python, PyTorch, HuggingFace Transformers, Tokenizers, Pandas, NumPy	Source Code
CNN based Image Analysis: Detecting Defects in Plant Leaf - Trained CNN architectures on labeled datasets to classify plant disease 14+ categories and improve early detection accuracy. - Integrated preprocessing, augmentation, and model evaluation pipelines for robust performance across varying images. Tech: Python, TensorFlow/Keras, NumPy, Matplotlib	Source Code
AI based OCR: Document Processing for Invoice Summary Generation - Implemented OCR-based text extraction followed by automating financial document processing to reduce manual effort and improve accuracy in data extraction. Tech: React, JavaScript, Mindee OCR Image Parsing API, FastAPI	Source Code
LLM powered Multi-Agent System: Enterprise Knowledge Platform - Fabricated an assistant capable of answering enterprise knowledge queries using workflow with 4 agents. - Devised document ingestion, chunking, embedding generation, vector indexing, retrieval and answer synthesis pipeline. - Fabricated retrieval orchestration with ChromaDB and RAG, context-aware prompting and citation-aware responses. - Integrated LLMops practices including experiment tracking, prompt versioning, evaluation datasets, and automated regression testing for continuous improvement of agent performance. Tech: Python, PineconeDB, Ollama, DeepSeek	Source Code
Automated ML Pipeline: Taxi Demand Prediction - Orchestrated an end-to-end MLOps pipeline to automate NYC taxi demand prediction with 1M+ records, integrating data ingestion, preprocessing, feature engineering across 10+ features, model training, evaluation, and artifact versioning. - Orchestrated monthly workflows using Apache Airflow and enacted CI/CD with GitHub Actions for automated model retraining, validation, and deployment-ready pipelines. Tech: Python, Apache Airflow, Scikit-learn, Pandas, MLflow, MinIO, GitHub Actions, Docker	Source Code
Airport Authority of India, Kalaburagi: Management Information System - Developed a web-based internal management platform for the Airport Authority of India (AAI), streamlining operations across 5+ departments. - Developed User Management and Daily Task Management modules for admins to manage 500+ staff profiles at different levels, attendance, and department-specific task assignment/verification workflows. - Effected an internal Messaging module and a BCAS Compliance Dashboard unifying safety inspections, drills, and regulatory records; Tech: ReactJS with Vite, Tailwind, Postman.	Source Code
Task-Specific Personal Assistant using Agentic AI In Progress Built an agent-based AI assistant capable of executing task-specific workflows using LLM reasoning and tool integration.	Source Code

- Enacted multi-step decision-making pipelines with dynamic prompt chaining and context management.
- Enabled automation of tasks such as query handling, summarization, and structured output generation.
- Integrated RAG based concepts to produce content more relevant to the conversation.
- **Tech:** Python, CrewAI, Ollama, DeepSeek, FastAPI, ChromaDB

Restaurant Management System

[Source Code](#)

- Engineered a full-stack restaurant management system with a strong focus on database design, enabling real-time handling of orders, inventory, and customer data.
- Conceived highly normalized relational schemas in PostgreSQL, leveraging indexing, joins, and query optimization techniques to ensure efficient data retrieval and transactional consistency.
- **Tech:** ReactJS, Postgres, NodeJS

Multi-Agent AI System: Fault Detection in Electrical Appliances

- Created a vector database from appliance manuals using ChromaDB and executed Retrieval-Augmented Generation (RAG) to extract operational thresholds and parameter ranges.
- Compared real-time sensor readings against retrieved reference values to detect anomalies and identify potential faults.
- Modeled 4 Agents using LLM and LangChain and LangGraph to automate reasoning, validation, and fault diagnosis.
- **Tech:** LangChain, LangGraph, ChromaDB (Vector DB, RAG), DeepSeek (LLM)

RESEARCH, PUBLICATIONS AND PATENTS

Generative AI-Enhanced Data Marketplaces: Algorithms and Platforms for Secure Data Exchange

[Published](#)

- **Patent Application pending** - Application No: 202541123233
- Presented at the **4th IEEE International Conference on Sensors and Related Networks SENNET-2025 Special Focus on Digital Healthcare** on the 26th of July, 2025.
- Created a web marketplace enabling users to generate, share, and access AI-enhanced datasets, automating dataset creation with generative models to improve data diversity and ML project quality.
- Tech Stack: JavaScript, TensorFlow, ReactJS, NodeJS, FAISS

Comparative Analysis of Fine-Tuned Convolutional Models in Detecting Face Masks

[Source Code](#)

- Compared performance of 5 industry level CNN models for face-mask detection using fine-tuning on 14000+ image data.
- Designed preprocessing, augmentation and transfer learning pipelines, evaluating models using accuracy, precision, recall, F1-score and confusion matrices to identify the most effective CNN architecture for real-world mask detection.

CERTIFICATION

Mastering GPU Parallel Programming with CUDA - Udemy

[Verify](#)

Introduction to Model Context Protocol (MCP) - Anthropic

[Verify](#)

Apache Spark Programming in Scala - Udemy

[Verify](#)

Generative AI Fundamentals - IBM

[Verify](#)

Supervised Machine Learning: Regression and Classification - Coursera & DeepLearning.AI

[Verify](#)

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R

Frameworks: React, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI

Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse

Libraries: pandas, NumPy, Matplotlib